1. Construct the phylogenetic tree

There are 16 possible changes from one nucleotide to the other. The changes can be represented as 4X4 transition matrix.

$$\mathbf{P} = \begin{array}{cc} \begin{array}{c} A \\ C \\ G \\ T \end{array} & \begin{bmatrix} 0.976 & 0.01 & 0.007 & 0.007 \\ 0.002 & 0.983 & 0.005 & 0.01 \\ 0.003 & 0.01 & 0.979 & 0.007 \\ 0.002 & 0.013 & 0.005 & 0.979 \end{bmatrix} \\ & \begin{array}{cccc} A & \quad C & \quad G & \quad T \end{array} \end{array}$$

Please write the equation and calculate the log likelihood for the following tree topologies. Then decide the topology for the phylogenetic tree by maximum likelihood.

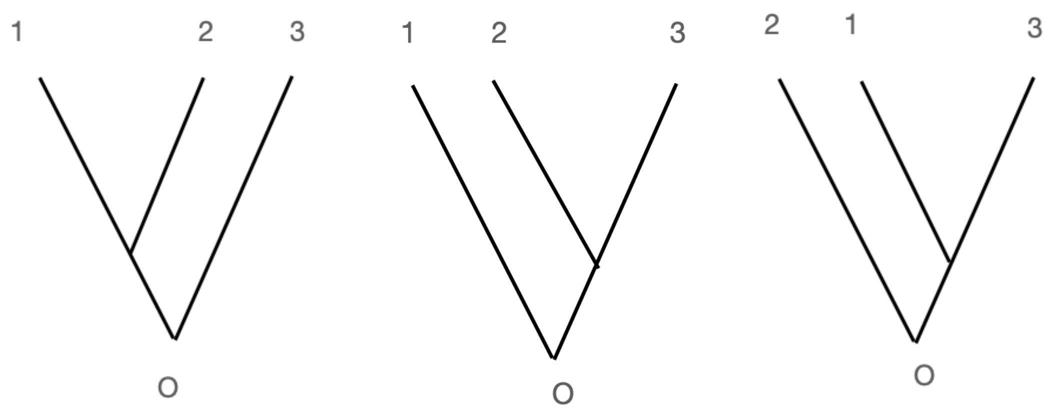
Suppose the sequences are as follows

X0: AAA

X1:AGT

X2:AAC

X3:AGG




2. Please download the gene expression table from iSpace.

1). Apply cell type cluster for the data. (Please store the cluster label for each cell);

2). For the two clusters that have most cells (summary the numbers as table), detect the differential expressed genes across the two cluster. Remember to conduct multiple test adjustment. (Please store pvalue, and fold change for each gene; Alpha=0.05, fold change >2 as threshold)

3). For the detected differential expressed genes, apply the gene set enrichment to see what is the most relative biological function.

(Please indicate method you used in each step and store the result as ".csv" file)

3. See following description about detecting differential expressed gene by linear regression model.

To determine whether the read count differences between different conditions for a given gene are greater than expected by chance, DGE tools must find a way to estimate that difference using the information from the replicates of each condition. `edgeR` (Robinson et al., 2010), `DESeq/DESeq2` (Anders and Huber, 2010; Love et al., 2014), and `limma-voom` (Ritchie et al., 2015) all use regression models that are applied to every single gene. Linear regression models usually take the following typical form: $Y = b_0 + b_1 * x + e$ (Figure 24).

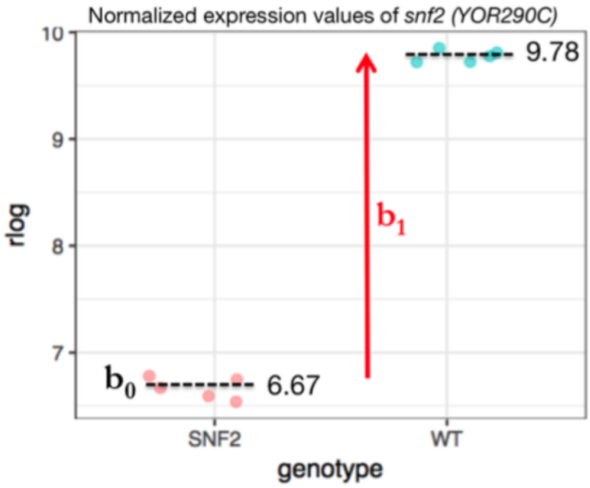


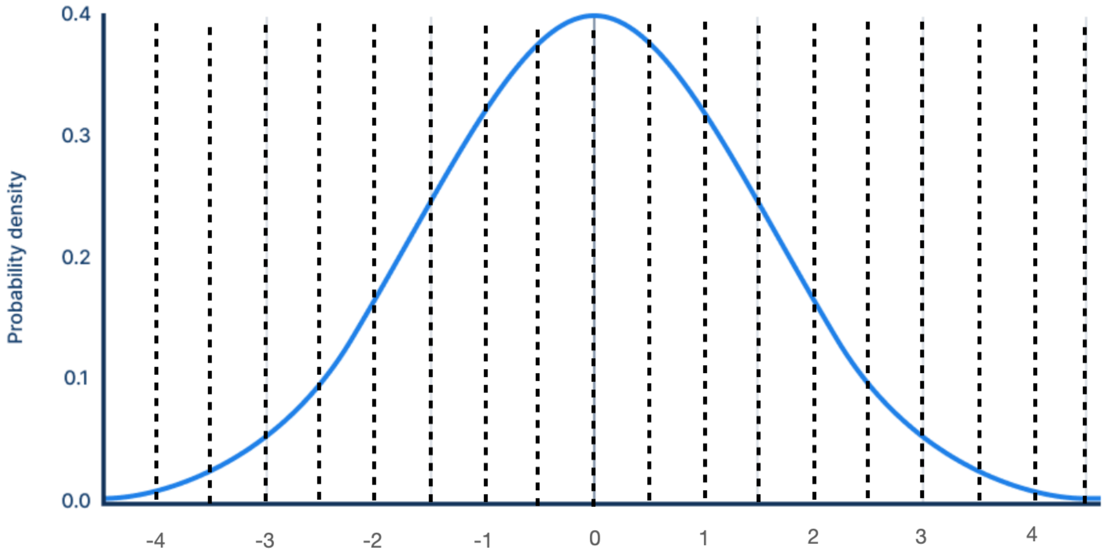

**Figure 24:** For the most basic comparison of two conditions, all (normalized) expression values, $Y$, can be described with the help of a regression model where $b_0$ represents the average value of the baseline group and $b_1$ the *difference* between baseline and non-reference group: $Y = b_0 + b_1 * x + e$.
$x$ is a discrete parameter, which is set to 0 for expression values measured in the baseline condition (here: SNF2) and set to 1 for the non-reference group (here: WT).

Here, $Y$ will entail *all* read counts (from all conditions) for a given gene. $b_0$ is called the *intercept*; $x$ is the condition (for RNA-seq, this is very often a discrete factor, e.g., "WT" or "mutant", or, in mathematical terms, 0 or 1), $e$ is a term capturing the error or uncertainty, and $b_1$ is the coefficient that captures the difference between the samples of different conditions*.

Given the normalized genes expression value as follow:

| Gene_1 | Gene_2 | Gene_3 | Gene_4 | Gene_5 | Gene_6 |
|---|---|---|---|---|---|
| T1: 4.1<br>T2: 5.2<br>T3: 2.8<br>C1: 5.6<br>C2: 3.7<br>C3: 3.8 | T1: 2.6<br>T2: 3.9<br>T3: 1.7<br>C1: 6.0<br>C2: 2.1<br>C3: 4.8 | T1: 3.3<br>T2: 2.5<br>T3: 3.0<br>C1: 5.3<br>C2: 3.2<br>C3: 5.0 | T1: 1.7<br>T2: 2.7<br>T3: 2.5<br>C1: 0.8<br>C2: 1.0<br>C3: 2.3 | T1: 2.8<br>T2: 3.0<br>T3: 1.9<br>C1: 3.3<br>C2: 3.2<br>C3: 2.7 | T1: 5.5<br>T2: 4.7<br>T3: 6.1<br>C1: 3.0<br>C2: 2.8<br>C3: 1.4 |

There are two groups (T and C). We need to determine whether the expression differences between different conditions for a given gene are greater than expected by chance. Here we use simple linear regression model to fit the difference between different conditions. Calculate the b0 and b1 for each gene.
Suppose the distribution of b1 for all genes are as follows,




Which genes are differential expressed genes, if we take 10% level of significance (alpha=0.1) ?